Vadim Fedorov

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EDUCATION

University of California, Davis *Expected June 2028*
B.S. Electrical Engineering | GPA 3.54 | Dean's List: Fall 2025, Winter 2025

RESEARCH AND TECHNICAL EXPERIENCE

Undergraduate Robotics Researcher *Jan 2026 - Present*
Laboratory for AI, Robotics, and Automation (LARA) | UC Davis

- Diagnosed I2C bus contention behind two-second teleoperation crashes in a 19-actuator robotic hand; rewrote firmware on both signal paths to sustain control at approximately 50 ms end-to-end latency.
- Designed and tested custom control PCBs from schematic through bring-up, adding I2C pull-ups and dedicated voltage regulation.
- Traced limited simultaneous motion to a shared power feed through the PCB stack, then designed a separate parallel power distribution board in KiCad; verified all motors at full PWM without freezes or motion lag.
- Integrated two wrist rotary actuators into a Manus glove, PyBullet, and micro-ROS teleoperation pipeline, extending control from hand joints to wrist articulation.

Neurotech Club, Projects Division *Dec 2025 - Jun 2026*
BCI Prosthetic Leg Control | OpenBCI Cyton and Arduino Mega test rig

- Designed and ran a cued ankle motor-imagery paradigm, collecting labeled MOVE/REST EEG trials with an eight-channel OpenBCI Cyton for a prosthetic-leg test rig.
- Built the MNE pipeline for XDF import, 60 Hz notch and 8-30 Hz bandpass filtering, common-average referencing, artifact rejection, epoching, and CSP/LDA decoding.
- Linked live BrainFlow decoding to an Arduino Mega actuator trigger; added dry-run, decision thresholds, refractory timing, and REST-output blocking; checked replay and session holdout.

SELECTED PROJECTS

PyBCI Live EEG Interface *Jan 2026 - Present*
Open-source contribution to EEG research software

- Built the live EEG acquisition and streaming backend, connecting board data to the research interface.
- Implemented stacked real-time EEG plots and start, pause, save, and stop recording controls in the frontend.

ADDITIONAL EXPERIENCE

Customer Service Representative *Jun 2024 - Sep 2025*
Cherry Chase Pet Hospital | Sunnyvale, CA

- Managed client communication across phone, email, and front desk; coordinated appointments and intake with veterinary staff during peak clinic hours.

PUBLICATION

The Implementation of Ethical AI Within Autonomous Vehicles *2024*
Journal of Student Research

TECHNICAL SKILLS

Programming and firmware: C/C++, embedded C, Python, Java, MATLAB
Electronics and CAD: KiCad, SolidWorks, PCB schematic and layout, board bring-up, I2C, Teensy, soldering
Robotics and neurotech: micro-ROS, PyBullet, MNE, BrainFlow, OpenBCI Cyton, CSP/LDA
Tools: Git, Linux, prototyping